

Deep Learning for Natural Language Processing

Assignment- 2 Solutions

TYPE OF QUESTION: MCQ (Single Correct)

Number of questions: 10

Total marks: $10 * 1 = 10$

1.

A shallow ReLU network has hidden pre-activations $z = Wx + b$ with $W = \begin{bmatrix} 1 & -1 \\ 2 & 1 \\ -1 & 1 \end{bmatrix}$ and $b = [0, -3, 1]$. The hidden activations $h = \text{ReLU}(z)$ feed an output $y = v^T h$ with $v = [1, 2, 1]$ and no output bias. For the input $x = [1, 2]$, what is y ?

(A) 0

(B) 2

(C) 4

(D) 6

Answer: (C)

Solution: Pre-activations: $z_1 = 1(1) + (-1)(2) + 0 = -1$; $z_2 = 2(1) + 1(2) - 3 = 1$; $z_3 = -1(1) + 1(2) + 1 = 2$.

After ReLU: $h = [0, 1, 2]$ (the first unit is switched off).

Output: $y = 1(0) + 2(1) + 1(2) = 4$.

2.

A single hidden unit computes $h = \text{ReLU}(wx + b)$ with $w = -2$, $b = 1$, evaluated at $x = 3$. What is $\partial h / \partial w$ at this input?

(A) 3

(B) 0

(C) -2

(D) -6

Answer: (B)

Solution: Pre-activation = $w x + b = (-2)(3) + 1 = -5 < 0$, so the ReLU is in its flat region and $h = 0$.

The derivative of ReLU is 0 for negative pre-activations, hence $\partial h / \partial w = 0 \times x = 0$.

This is the “dead ReLU” situation: no gradient flows back through this unit for this input.

3.

A softmax output layer receives the logits [2, 1, 0]. If the correct class is the first one, what is the cross-entropy loss (using natural logarithm)?

(A) 0.335

(B) 0.408

(C) 0.665

(D) 1.407

Answer: (B)

Solution: $\exp(2) = 7.389$, $\exp(1) = 2.718$, $\exp(0) = 1.000$; sum = 11.107.

Softmax probabilities = [0.665, 0.245, 0.090].

Loss = $-\ln(0.665) = 0.408$. Option C is the probability itself, not the loss.

4.

A network with a single weight w produces the output $y = w^3$, and the loss is $L = \frac{1}{2}(y - t)^2$ with target $t = 5$. If the current value is $w = 2$, what is $\partial L / \partial w$?

(A) 12

(B) 24

(C) 36

(D) 108

Answer: (A)

Solution: $y = w^3 = 8$.

$$\partial L / \partial y = y - t = 8 - 5 = 3.$$

$$\partial y / \partial w = 3w^2 = 3(4) = 12.$$

By the chain rule, $\partial L / \partial w = 3 \times 12 = 36$.

5.

Gradient descent is run on $L(w) = w^2 - 4w + 3$ starting from $w_0 = 4$ with learning rate $\eta = 0.1$. What is the value of w after TWO updates?

(A) 3.28

(B) 3.60

(C) 3.40

(D) 2.21

Answer: (A)

Solution: $\partial L / \partial w = 2w - 4$.

Step 1: gradient = $2(4) - 4 = 4$, so $w_1 = 4 - 0.1(4) = 3.6$.

Step 2: gradient = $2(3.6) - 4 = 3.2$, so $w_2 = 3.6 - 0.1(3.2) = 3.28$.

(The minimum lies at $w = 2$, so the iterates are converging towards it.)

6.

A model is trained on 50,000 examples using mini-batch stochastic gradient descent with a batch size of 250, for 12 epochs. How many parameter updates are performed in total?

(A) 200

(B) 600

(C) 2400

(D) 600000

Answer: (C)

Solution: Updates per epoch = $50,000 / 250 = 200$.

Total updates = $200 \times 12 = 2,400$.

Full-batch gradient descent would perform only 12 updates on the same budget of data passes.

7.

For the function $f(w_1, w_2) = w_1^2 - w_2^2$, the gradient vanishes at the origin. What kind of stationary point is the origin?

(A) A global minimum

(B) A local maximum

(C) A local minimum that is not global

(D) A saddle point

Answer: (D)

Solution: Along the w_1 -axis, $f = w_1^2$ has a minimum at 0; along the w_2 -axis, $f = -w_2^2$ has a maximum at 0.

The Hessian is $\text{diag}(2, -2)$, with eigenvalues of opposite sign, so the origin is a saddle point.

Plain gradient descent can stall near such points because the gradient magnitude becomes very small.

8.

Using the least squares loss $L = (1/N) \sum_i (y_i - \hat{y}_i)^2$ with $\hat{y} = wx + b$, evaluate the loss for $x = [1, 2, 3, 4]$, $y = [3, 5, 7, 10]$, $w = 2$ and $b = 1$.

(A) 0.00

(B) 0.25

(C) 0.50

(D) 1.00

Answer: (B)

Solution: Predictions = [3, 5, 7, 9]; residuals = [0, 0, 0, 1].

Sum of squared residuals = 1; dividing by $N = 4$ gives $L = 0.25$.

Option D is the sum of squared errors without the $1/N$ factor.

9.

A linear regression model predicts 3 real-valued outputs from 100 input features, with a separate bias for each output. How many parameters must be learned?

(A) 303

(B) 103

(C) 300

(D) 309

Answer: (A)

Solution: The weight matrix has $100 \times 3 = 300$ entries and there are 3 bias terms.

Total = $300 + 3 = 303$ parameters.

This is exactly a shallow network with no hidden layer and identity activation.

10.

Which of the following statements is FALSE?

(A) A shallow network with sufficiently many hidden units can approximate any continuous function on a compact subset of $\mathbb{R}^D$ to arbitrary precision.

(B) The universal approximation theorem guarantees that gradient descent will find the parameters achieving that approximation.

(C) Increasing the number of hidden units increases the number of linear regions a ReLU network can represent.

(D) The universal approximation theorem gives no practical bound on how many hidden units are required for a given function.

Answer: (B)

Solution: The universal approximation theorem is an existence (representational) result: it says a suitable set of parameters exists, but says nothing about whether an optimisation algorithm such as gradient descent can actually find them. Hence B is false.

A, C and D are correct statements about the scope and limitations of the theorem.
